IT4010 Research Project

How to Prepare for PP2

Evaluation Strategy • Demo Planning • Rubric Alignment • AI Policy

6 minutes briefing + 30 minutes demo/Q&A

Today’s Two-Hour Goal

By the end, every team should know

1. How to structure the 1-minute video and 5-minute briefing
2. What examiners look for in the 30-minute demo/Q&A
3. How the rubric translates into observable evidence
4. How to defend research, development, ethics, and AI use

The mindset shift

Do not ask evaluators to believe that your project is good. Give them enough evidence that they can reach that conclusion themselves.

Immediate outcome after today

A revised PP2 deck, rehearsed demo story, evidence folder, and Q&A preparation plan.

PP2 Format: Control the Time or Time Will Control You

1-minute video

Show the problem, solution idea, and impact quickly. It should create context, not replace the demo.

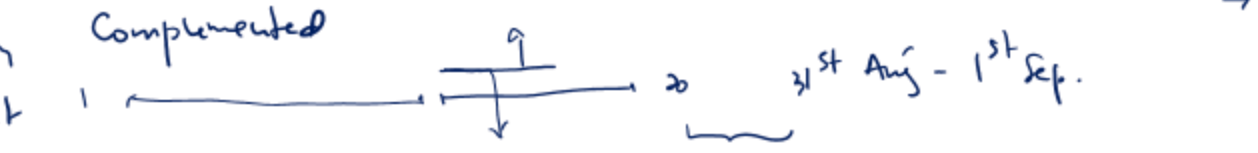
5-minute presentation

A panel-selected member gives the overview. Everyone must therefore know the whole project.

30-minute demo + Q&A

Demonstrate one integrated solution, individual contributions, evidence, and defensibility.

What students must avoid: long introductions, disconnected module demos, searching for files during the demo, and saying "we planned to finish this later."

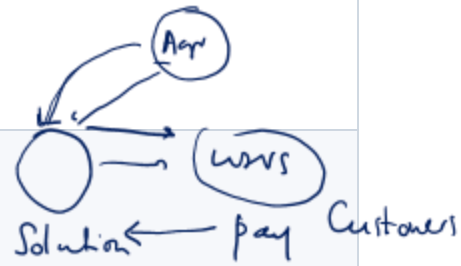






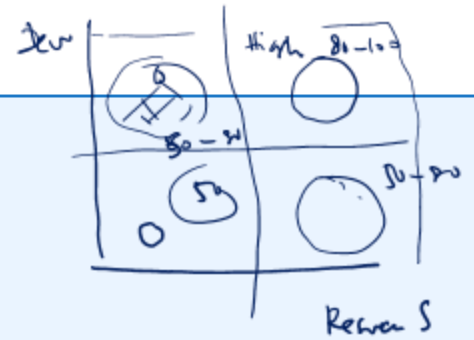
How PP2 Is Marked: What Gets Rewarded

Rubric area	Weight	What the panel needs to see
Proven gap / creative solution	10%	Clear problem + prototype transformed into current implementation
Capability in stream knowledge	20%	Specialized knowledge and technologies applied with depth
Solution implementation	40%	Design, completion, standards, PLSSE, NFRs
Communication	15%	Structure, Q&A, stage presence, time control
Commercialization / entrepreneurship	15%	Evidence of business potential and user benefits



The 40% solution implementation section is the largest single mark zone: design, completion, standards, ethical/security/social issues, and NFRs.

PP2 Must Show Both RP Strands



Research strand

Research gap
Objectives
Methodology
Baseline / comparison
Metrics
Results and limitations

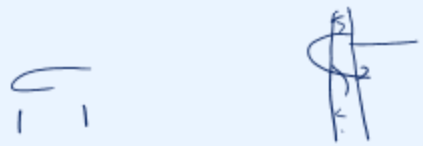
Development strand

Architecture
Integrated components
Working functions
Testing
NFRs
Deployment / demo readiness

High marks come when the two strands support each other: the system generates the evidence, and the evidence proves the research claim.

The 1-Minute Video: Create Context, Not Confusion

Suggested 60-second structure



0–10s: real problem and affected users
10–25s: what your solution does
25–45s: one end-to-end workflow or result
45–55s: impact / value proposition
55–60s: transition to live presentation

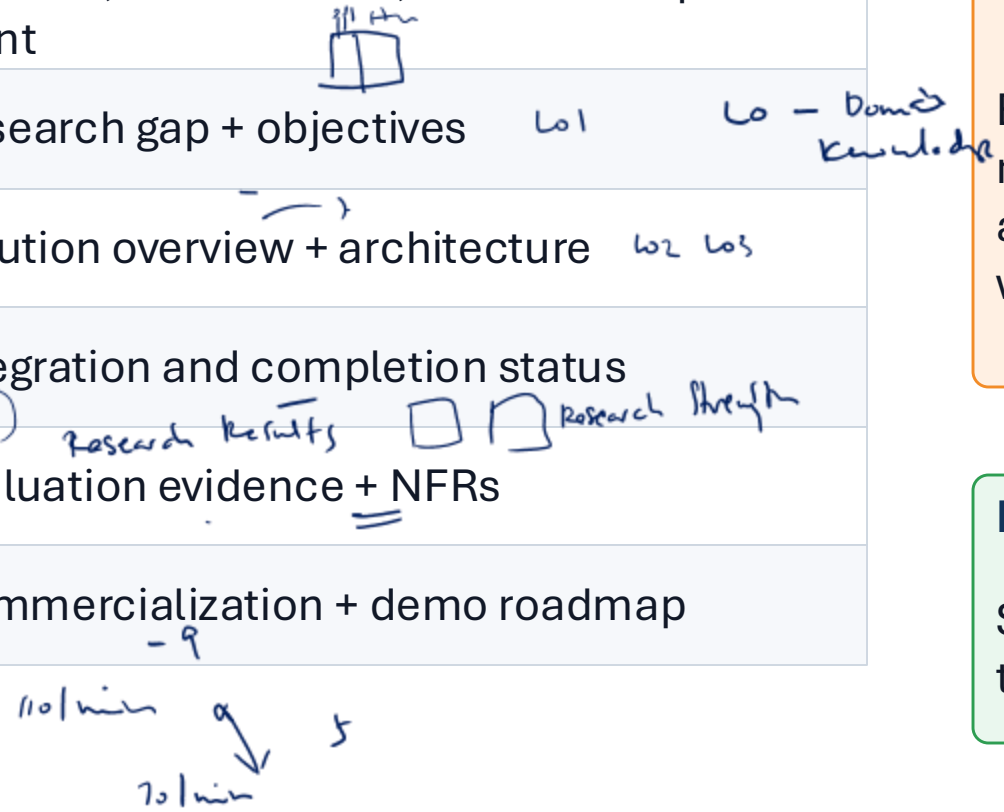
Do not use the video to...

Explain theory
Show source code
Add unrelated animations
Hide a weak live demo
Use copyrighted/AI-generated media without disclosure when required

Video should be professional, short, and directly connected to the problem-solution story.

The 5-Minute Presentation: One Member, Whole Team Prepared

Time	What to cover
0:00–0:45	Problem, stakeholders, measurable pain point
0:45–1:30	Research gap + objectives
1:30–2:30	Solution overview + architecture
2:30–3:30	Integration and completion status
3:30–4:30	Evaluation evidence + NFRs
4:30–5:00	Commercialization + demo roadmap



Critical reminder

Evaluators select the presenter. Every member must be ready to give this briefing and answer initial questions about the whole project.

Presentation goal

Set up the demo so the panel knows what to observe and how to judge it.

The 30-Minute Demo/Q&A: Demonstrate One Integrated Solution

Start with end-to-end story

Show the main user workflow first. Do not begin with isolated technical modules.

Then show component depth

Each student demonstrates their contribution, integration points, and technical decisions.

Finish with evidence

Show evaluation, testing, NFRs, Git/Planner evidence, user feedback, and limitations.





- Use test data prepared in advance.
- Prepare a backup recording/screenshots only as a contingency.
- Do not show source code unless the panel asks.
- Keep the transition between members smooth.
- Allocate time for questions during the demo.

System Maturity Checklist for PP2

- All four components integrated
- Core workflows run end-to-end
- Major functional requirements completed
- Important NFRs demonstrated
- No major runtime errors
- Deployment/demo setup ready
- Test data and backup plan ready

Completion is judged against the approved plan — not against what the team wishes to show on the day.

Research Maturity Checklist for PP2

- Research gap and objectives restated clearly
- Evaluation aligned with objectives
- Baseline or comparison strategy shown
- Metrics justified and results available
- Limitations and validity threats acknowledged
- Claims supported by data, not opinions
- AI use disclosed where relevant

A complete system without defensible evidence is still weak research.

Rubric Decoded: Proven Gap / Creative Solution (10%)

What to show

Problem definition must refer to the current implementation. Show why the problem still matters and how the solution responds to it.

Prototype → implementation

Explain how the original prototype evolved into functional units, modules, workflows, and evidence-producing features.

Strong wording

“Our proposal prototype had A and B. In the current implementation these became modules X and Y, integrated through API Z and validated using metric M.”

Rubric Decoded: Capability in Stream Knowledge (20%)

Stream evidence	What panels expect
AI/Data Science	Model choice, dataset handling, metrics, baseline, validity
Software Engineering	Architecture, design patterns, testing, deployment, maintainability
Cyber Security	Threat model, secure coding, test cases, evidence of controls
Interactive/UX systems	User-centred design, usability testing, accessibility, adoption evidence



Do not merely name technologies. Demonstrate why you selected them, how you used them, and what evidence shows they worked.

Rubric Decoded: Solution Implementation (40%)

Design Excellence 30%

Architecture, modularity, scalability, security-aware design, integration logic.

Completion 40%

~90% completion where applicable, no major delays, corrective actions if needed.

Standards 20%

Git, Planner, testing, coding standards, documentation, deployment evidence.

PLSSE + NFR 10%

Professional, legal, social, security, ethical issues + NFR evidence.

This is the largest mark category. The panel must see disciplined engineering, not just working screens.

Show architecture before features.
Show integration evidence before isolated modules.
Show standards through artifacts: commits, tests, documentation, Planner, logs.
Show NFRs through measurements: response time, usability, reliability, security checks.

Lean Canvas → 2
VPC

Commercialization / Entrepreneurship (15%): Evidence, Not Wishes

Who benefits?

Target users, buyers, decision makers, and beneficiaries.

Why will they adopt?

Value proposition: time, cost, accuracy, safety, convenience, compliance, access.

Can it be sustained?

Cost estimate, deployment plan, support, scalability, pilot strategy.

What evidence?

User feedback, market comparison, measured benefits, stakeholder validation.

Custome ← Validation

A commercial story is strongest when it is connected to user benefits and evaluation results.

Avoid: “This can be sold to everyone.”
Prefer: “Our initial users are X. They currently face Y. Our solution reduces/ improves Z, shown by evidence E.”

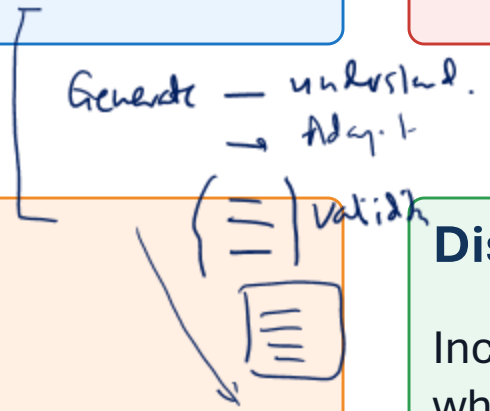
AI Policy at PP2: Use Responsibly, Disclose Honestly

Allowed with responsibility

Literature exploration, code assistance, drafting support, debugging ideas, model/tool use — when verified and disclosed as required.

Not acceptable

Fabricated results, fake citations, undisclosed AI-generated work presented as own, unverified code, unapproved use of sensitive data in AI tools.



What panels may ask

- Which AI tools did you use?
- What did you verify yourself?
- How did AI influence code/model/data/report?
- Can you explain and defend the output?

Disclosure line

Include a short AI-use disclosure in report/slides where applicable. Responsibility for accuracy, ethics and final outcome remains with students.

Professional, Legal, Social, Security & Ethical Issues

Issue type	What to identify at PP2
Professional	Responsibilities, limitations, user training, maintenance
Legal	Data protection, consent, licences, compliance where applicable
Social	Who is affected, who may be excluded, unintended consequences
Security	Authentication, authorization, privacy, audit logs, data protection
Ethical	Bias, fairness, transparency, harm, human oversight

Do not use generic slides. Show the issues specific to your system and your mitigation actions.

Prepare Your Evidence Pack Before PP2

Development evidence

GitHub commits
Branch/PR history
MS Planner tasks
Integration logs
Deployment notes

Testing evidence

Unit tests
Integration tests
System tests
NFR measurements
Bug fixes/regression notes

Research evidence

Dataset/test cases
Baseline results
Metrics
Graphs/tables
User feedback
Limitations

Governance evidence

AI disclosure
Ethics/privacy notes
Licences
Security controls
Supervisor feedback

Evidence should be ready before the demo — not searched for while the panel is watching.

Weak PP2 vs Strong PP2

Weak case

Four separate modules
Vague claims
No baseline
No NFR evidence
One member dominates
AI/tools not disclosed
Commercial value unsupported

Strong case

One integrated solution
Claims backed by metrics
Baseline comparison
NFRs measured
Every member ready
AI use transparent
User benefits evidenced

Ask students: “Which side does your team resemble today?”

Q&A Defence: Questions Behind the Questions

Panel question behind the question	Your answer must show
Does it work?	Show stable integrated workflow.
Did you build it properly?	Explain architecture, design decisions, standards.
What did each member do?	Show ownership, code evidence, component depth.
What is the research evidence?	Explain metrics, baseline, results, limitations.
Can it create value?	Show user benefits, market, feasibility.
Was AI used responsibly?	Disclose tools, verify outputs, defend decisions.

Selected Presenter Rule: Positive Preparation for Industry

What will happen

The 5-minute presentation may be delivered by a team member selected by evaluators.

Why this is fair

Every member receives marks and must understand the full solution, not only one isolated module.

Why this helps you

Interviews ask about the whole project. Strong graduates can explain architecture, integration, methods, and impact.

Team preparation rule

Each member rehearses: problem, gap, objectives, architecture, all components, their own contribution, evaluation, and commercialization.

48-Hour Rehearsal Plan

Rehearsal 1

Run full video + 5-min briefing + demo without interruption. Identify timing problems.

Rehearsal 2

Switch presenter. Run again. Ask hard questions after every demo segment.

Rehearsal 3

Technical rehearsal with same laptop/projector/network/power setup. Prepare backup.

Final check

Evidence folder, test data, accounts, extension cable, HDMI, chargers, offline backup.

In-Class Activity: PP2 Readiness Scorecard

Area	Score 1–5	Immediate action
System integration		
Completion vs plan		
Design explanation		
Standards/best practices evidence		
NFR evidence		
Research evaluation evidence		
Individual ownership		
AI policy / ethics disclosure		

Any score of 1 or 2 becomes a compulsory action item before PP2.

Final PP2 Checklist

- [ ] 1-minute video ready and tested
- [ ] 5-minute briefing rehearsed by all members
- [ ] Integrated demo flow rehearsed
- [ ] Evidence pack ready
- [ ] Rubric alignment checked
- [ ] AI use disclosure prepared
- [ ] User/commercialization evidence included
- [ ] Backup plan ready
- [ ] Q&A practiced
- [ ] All members understand full system

Do not arrive at PP2 hoping the system will work. Arrive knowing it will work because you rehearsed it.

Final Message

PP2 is not a claim that your project is good.
It is your opportunity to prove it.

Integration + Evidence + Ownership + Professionalism = PP2 Readiness

Go in prepared. Demonstrate clearly. Defend honestly.